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The impact of international student education on facilitating global trade: A case study of chinese education for foreign students in China

Chengang Zeng [a], Jing Guo [b,*], Yuxun Zheng [b]

[a] School of International Chinese Studies/Research Center for Global Chinese Language Development and the Joint Digital Intelligence Laboratory of International Chinese Language Education, East China Normal University, Shanghai, PR China
[b] College of International Chinese Studies, Beijing Language and Culture University, Beijing, PR China

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ABSTRACT

As China has become the third largest destination for international students globally and the largest in Asia, assessing the social benefits of its education for foreign students is crucial for formulating and optimizing relevant education policies. This study explores the role of educating international students in China in fostering the country's foreign trade. The regression analysis results indicate that such education significantly promotes bilateral trade between China and foreign students' native countries. Meanwhile, the impact on export trade is more pronounced than on import trade. Additionally, various categories of international students in China contribute to bilateral trade, albeit with varying degrees of influence. Non-degree (government-sponsored) students exhibit a stronger impact than degree-seeking (self-funded) students. Moreover, the trade-enhancing effect of educating international students in China diminishes over time, indicating a declining trend. These insights provide valuable implications for China to strategically shape its international education policies and enhance its international trade relations.

1. Introduction

The interplay between education and economic development is a key research topic in the economics of education, characterized by a dual-dimensional analytical framework: At the individual level, education serves as a strategic investment in human capital, generating private economic returns by enhancing employability and income premiums (Schultz, 1963; Spence, 1974). At the national level, the educational system functions as a mechanism for nurturing workforce competitiveness, thereby driving comparative advantages in global trade. This study concentrates on the economic value of education at the macroeconomic level, particularly examining the role of international education in Chinese universities (i.e., foreign student enrollment in Chinese higher education institutions) in fostering the nation's foreign trade. This theoretical demarcation seeks to address an understudied tension in extant scholarship: the interplay between human capital accumulation and institutional knowledge transfer.

Educational economics has examined foreign language education's facilitating role in international trade. Language differences between countries can increase communication costs, thereby hindering international trade (Fidrmuc, Fidrmuc, 2016; Kioko, 2018). Foreign language education mitigates the negative impact of language barriers on international trade, enabling parties who share a common language to reduce the costs associated with information gathering, negotiation, and supervision (Wang and Zeng, 2022). Although some studies support the facilitating effect of foreign language education on international trade, they consistently indicate that this effect is relatively limited (Gould, 1994; Selmier, Oh, 2013; Toubal et al., 2017). Thus, education's role in promoting international trade is not solely related to foreign language education. Rather, other types of education also positively affect international trade.

Subsequently, educational economics has underscored the promotional effect of international education on international trade, including foreign language education's role as international students studying in the host country typically require language proficiency. International education fosters trade enhancement through three key mechanisms: First, immersive cultural adaptation grants students a nuanced understanding of destination-market regulations and consumer behaviors. They acquire an in-depth knowledge of local culture, institutions, and markets during their stay in China that is disseminated through social networks. This can help streamline trade by dismantling informal barriers, reducing transaction costs, and fostering bilateral trade growth

* Corresponding author.
E-mail addresses: cgzeng@chinese.ecnu.edu.cn (C. Zeng), guojing2021@blcu.edu.cn (J. Guo).

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(Murat, 2014). Second, prolonged exposure to the host-country's business environment cultivates institutional knowledge beyond language skills. Immigrants, as highlighted by Borgatti and Cross (2003), mitigate information asymmetry and diversify information sources. This helps reduce firms' costs for accurate data, thereby reducing trade expenses and stimulating foreign trade. Third, transnational networks forged during student exchanges establish enduring trade partnerships, with returning students often bridging political and economic exchanges between their home countries and China after completing their studies in China (Murat, 2018). Furthermore, the long-term nature of international student education yields profound outcomes like technology transfer and investment collaboration. The short-term, instrumental focus of foreign language education may find it challenging to achieve it.

China's transition from a manufacturing-centric economy to a global innovation leader has cemented its status as the world's second-largest economy and the largest goods trader (World Bank, 2023). While tariff reductions have advanced traditional trade liberalization, persistent structural frictions, particularly intangible barriers stemming from cultural norms and regulatory disparities, continue imposing significant transaction costs on cross-border economic activities (Anderson and Van Wincoop, 2004; Baldwin, 2016). International education has emerged as an under-explored yet transformative mechanism for alleviating such intangible barriers through cross-border human capital exchange (Docquier, Lodigiani, 2010).

This study adopts an institutional capital framework to analyze how educational exchanges, particularly degree-seeking students' mobility, facilitate bilateral trade by transferring knowledge across borders and aligning regulatory frameworks. It transcends conventional linguistic approaches by integrating systemic institutional learning mechanisms (Knight, 2012; Murat, 2018). We empirically examine the impact of international student education in China on the country's foreign trade using panel data from 169 countries (2003–2022). However, moving beyond a narrow focus on trade volume, we situate our analysis within the broader discourse on education for development. We contend that trade facilitation is not an end in itself but a critical mechanism for achieving broader developmental goals. By reducing transaction costs and building trust, the trade links strengthened through educational exchanges can contribute to inclusive growth. Specifically, they can help integrate small and medium-sized (SME) enterprises into global value chains, foster regional integration through deepened economic interdependence, and promote long-term cooperation on shared challenges such as sustainable development. Overall, this study explores how international student education in China can serve as a conduit not only for commercial exchange but also for meaningful socio-economic development partnership between China and other countries, particularly developing nations.

This study is structured as follows: Section 2 establishes the theoretical foundations using an institutional capital framework. Section 3 delineates the pathways through which international students in China promote Sino-foreign economic and trade cooperation. Section 4 presents our model and analyzes the results. Section 5 presents the conclusions of the study.

2. Literature review

The nexus between international education and trade is well-established in economics, and has been primarily explained through human capital and trade cost theories. Human capital theory frames education as an investment that enhances individual productivity and reduces labor market information asymmetry (Schultz, 1963; Spence, 1974). In an international context, this translates to students acquiring host-country-specific knowledge, forming a type of "institutional adaptive human capital" (Murphy et al., 2018). Concurrently, trade cost theory posits that international trade is hindered by both tangible (e.g., tariffs) and intangible barriers, such as information asymmetry and institutional differences (Anderson, Van Wincoop, 2004). Education, particularly studying abroad, can be a mechanism to lower these intangible costs by improving language skills and fostering cross-cultural understanding (Gould, 1994; Fidrmuc, Fidrmuc, 2016).

A significant research strand focuses on the role of migrant and diaspora networks in facilitating trade, with international students acting as a specific, high-potential cohort. These networks can reduce information search costs, build trust, and shape consumer preferences, thereby acting as a substitute for imperfect market information (Gould, 1994; Rauch and Trindade, 2002; Murat, 2014). Studies consistently find a positive, though often modest, correlation between student mobility and bilateral trade flows (e.g., Combes et al., 2005).

However, the literature exhibits three key limitations that this study seeks to address.

First, a short-term and linguistic bias exist. Many studies emphasize the immediate, language-focused benefits of education (Gould, 1994; Toubal et al., 2017). However, the deeper, long-term accumulation of institutional capital, such as the knowledge of regulatory frameworks, business norms, and institutional networks (Knight, 2012; Murat, 2018), remains underexplored. Education's role in transferring this systemic institutional knowledge, crucial for navigating trade in complex and highly regulated sectors, remains a relative blind spot.

Second, studies often make homogenizing assumptions. The distinct impacts of different categories of international students are frequently overlooked, such as degree-seeking versus non-degree students or government-sponsored versus self-funded cohorts. This aggregation masks important heterogeneity in how different educational experiences translate into trade facilitation. Crucially, the mechanisms and intensity of influence likely vary significantly across these groups. The extant literature fails to adequately capture this nuance.

Third, the dynamics have been neglect. Many analyses provide a static snapshot of the education-trade relationship, ignoring how its effect may temporally evolve. Factors like improving communication technologies or the natural dissipation of network effects suggest a potential decay in the influence of student networks (Tadesse and White, 2010). However, this temporal dimension is rarely empirically investigated.

Furthermore, despite China's dual status as the world's top trader and a major global education hub, rigorous, large-N studies situated specifically in the Chinese context remain scarce. The unique characteristics of China's trade structure and its international education strategy warrant a focused investigation.

Addressing these gaps, our contributions are threefold. First, we move beyond a narrow linguistic focus to construct a comprehensive tripartite model of "cost reduction–preference shaping–direct promotion", which explicitly incorporates the institutional capital perspective. Second, we employ an augmented gravity model with an instrumental variable (IV) to address endogeneity concerns: the number of students from a country studying in the U.S. We systematically test for heterogeneous effects across student types and a time-decay effect. Third, by utilizing an extensive panel dataset from 169 countries over the 2003–2022 period, we provide timely empirical evidence on the role of international student education in China's global trade network construction, thereby offering insights relevant to other emerging economies.

3. Hypothesis

Drawing from extant research, this study posits that international students studying in China facilitate bilateral trade through three primary pathways: cost reduction, preference shaping, and direct promotion. Crucially, each pathway holds significant implications for broader developmental outcomes. We highlight these alongside the core trade mechanisms. The specific pathways, including their developmental extensions, are illustrated in Fig. 1.

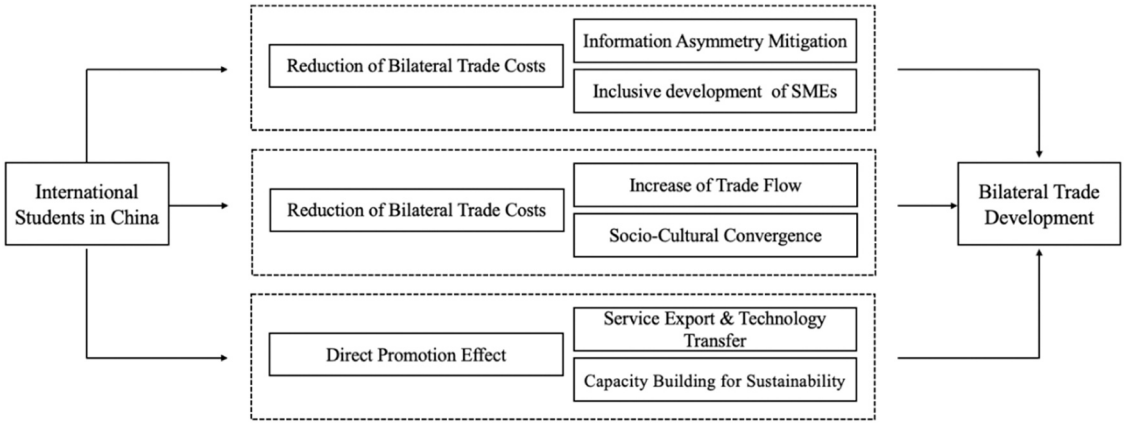


Fig. 1. Pathways for Promoting Sino-Foreign Economic and Trade Cooperation by International Students in China.

3.1. Cost reduction effect and inclusive market access

Hypothesis 1. (H1): The scale of international students in China positively affects bilateral trade volume by reducing transaction costs, an effect which may be particularly consequential for SME-led, inclusive growth.

The first pathway centers on reducing bilateral trade costs. International students act as conduits of localized information, mitigating information asymmetry, and overcoming linguistic and institutional barriers that typically hinder market entry. By leveraging their dual cultural and institutional knowledge, they can lower transaction costs for firms, acting as trusted intermediaries (Rauch and Trindade, 2002; Murat, 2014). This cost-lowering effect is hypothesized to be a primary driver of trade facilitation.

Crucially, this mechanism holds significant potential for inclusive development. High trade costs often disproportionately exclude SMEs from global value chains. By lowering these barriers, the networks formed by international students can enable SMEs in their home countries to access the Chinese market. This can foster a more diversified economic base and promote job creation, which is a key component of poverty reduction and resilient economic growth.

3.2. Preference shaping effect and socio-cultural convergence

Hypothesis 2. (H2): International student mobility between a partner country and China positively influences bilateral trade by shaping consumer preferences, a process that concurrently enhances social cohesion and cultural ties.

The second pathway involves shaping consumption preferences. International students naturally develop affinities for host-country goods and services during their stay. They often maintain these preferences after returning home (Gould, 1994). Concurrently, their presence in China stimulates interest in their home-country products. This two-way diffusion of preferences can directly increase trade flows in both directions.

Moreover, this pathway facilitates socio-cultural convergence and cohesion. The exchange of cultural products, from media and food to educational services, strengthens people-to-people bonds and fosters mutual understanding. This reduction in "cultural distance" (Felbermayr et al., 2010) builds a foundation of trust that is essential for long-term cooperation on regional and global challenges, thereby moving the relationship beyond transactional exchange toward stable partnership.

3.3. Direct promotion effect and capacity building for sustainability

Hypothesis 3. (H3): International student education in China directly promotes bilateral trade through service exports and technology transfer, with significant benefits for sustainable development and capacity building in partner countries.

The third pathway is direct trade promotion. International education is itself a significant export service under the GATS framework, directly contributing to trade volumes. It especially serves as a direct channel for knowledge and technology transfer, particularly in strategic sectors such as renewable energy, digital infrastructure, and public health.

This direct channel is intrinsically linked to capacity building and sustainable development. When international education involves training in advanced, sustainable technologies, it directly equips partner countries with the necessary skills and knowledge to advance their own development agendas. This transforms the relationship from simple commodity trade to an innovation partnership, directly contributing to sustainable development goals through technology transfer and human capital formation.

4. Design

4.1. Model, variables, and estimation strategy

4.1.1. Empirical model

We based our analysis on the structural gravity model of trade, which is the standard framework for analyzing bilateral trade flows. The baseline econometric specification is as follows:

$$LnTrade_{it} = \beta_0 + \beta_1 Ln(Student_{it}) + \beta_n Ln(\vec{x}_{it}) + \mu_i + \varepsilon_{it} \quad (1)$$

Here, Trade_it denotes China's bilateral trade volume, including imports and exports, with country i in year t. Student_it signifies the number of international students from country i enrolled in China during year t. It serves as a proxy for the progress of education for these students. Depending on the research context, Student_it can also signify the counts of degree-seeking students, non-degree students, students with Chinese government scholarships, and those without scholarships. x_it represents a cluster of control variables influencing bilateral trade between China and the source country. These primarily encompass population size (POP_it), foreign trade reliance (Depend_it), per capita GDP (Per_GDP_it), tax rate (Tax_rate_it), exchange rate (Exchange_rate_it), land mass (Area_it), and the direct distance from China (Distance_it). μ_i accounts for individual effects, whereas ε_it signifies random perturbations.

4.1.2. Data Sources

We collected data on bilateral trade from the National Bureau of Statistics of China[1] and the number of international students in China from the Ministry of Education's China Education Statistical Database,[2] which covers the period from 2003 to 2022. Data on control variables include population, per capita GDP, and exchange rates from the World Bank's Global Economic Indicators Database,[3] and land area, foreign trade dependence, and tax rates from the World Development Indicators (WDI) Database.[4] Straight-line distances between capitals and Beijing were manually calculated using Google Maps.[5] Data on the number of students studying in the US, which were used as an IV, were sourced from the Institute of International Education's Open Doors Report.[6] The panel dataset of 3380 observations across 169 countries was obtained after data cleaning, missing values interpolation, and the exclusion of countries with significant data gaps. Table 1 presents the variable definitions and descriptive statistics.

4.1.3. Estimation strategy and endogeneity addressing

Our estimation strategy was designed to ensure robust identification and address potential methodological issues.

First, we tested for multicollinearity. The Variance Inflation Factor (VIF) test showed a maximum value of 3.29 and mean of 2.01, well below common thresholds. Thus, multicollinearity is not a serious concern.

Endogeneity can arise from reverse causality (e.g., growing trade ties might encourage more students to study in China) and omitted variable bias. To mitigate this, we employed an IV approach. Following established literature (e.g., Javorcik et al., 2011; Murat, 2014), we used the number of students from country i studying in the U.S. as an instrument for the number of students from country i in China. Both the US and China are major destinations for international students. Global educational demand shocks can affect student flows to both countries, ensuring the instrument's relevance. However, the number of students in the US is unlikely to directly affect bilateral trade between China and their home country, satisfying the exclusion restriction.

The relevance of the instrument was strongly supported. The limited information maximum likelihood (LIML) test and a weak identification test yielding an F-statistic of 60.04, thereby comfortably rejecting the null hypothesis of a weak instrument.

To choose between fixed (FE) and random effects (RE) models within the IV framework, we proceeded as follows. The least squares dummy variable (LSDV) test indicated the presence of significant individual (country-specific) effects. We then estimated both two-stage least squares (2SLS) models with FE and RE. A Hausman test was subsequently conducted to compare the two consistent estimators under the null hypothesis. The test results favored the RE model. Hence, we adopted it for our baseline estimates. Finally, to account for potential heteroskedasticity, all reported standard errors are robust.

The panel dataset of 3380 observations across 169 countries was obtained after data cleaning, missing values interpolation, and the exclusion of countries with significant data gaps. The variable definitions and descriptive statistics are presented in Table 1.

Table 1
Descriptive Statistics.

	Variable Name	Definition	Mean (Standard Deviation)	Sample Size
Dependent Variable	$biatrade_{it}$	Total Bilateral Trade between China and Country i in Year t (10,000 USD)	1206618.0 (±4154962.0)	3380
	$Imoprt_{it}$	China's Total Imports from Country i in Year t (10,000 USD)	548704.8 (±1873262.0)	3380
	$Export_{it}$	China's Total Imports from Country i in Year t (10,000 USD)	657886.0 (±2543169.0)	3380
Independent Variable	$Student_{it}$	Number of Students from Country i Studying in China in Year t (persons)	1415.0 (±5124.0)	3380
	$Studre_{it}$	Number of Degree Students from Country i in China in Year t (persons)	590.2 (±2176.1)	3380
	$Stundre_{it}$	Number of Non-Degree Students from Country i in China in Year t (persons)	825.5 (±3287.7)	3380
	$Stuship_{it}$	Number of Students from Country i in China with Chinese Government Scholarships in Year t (persons)	127.9 (±353.3)	3380
	$Stunship_{it}$	Number of Students from Country i in China without Chinese Government Scholarships in Year t (Total Students from Country i in China minus Chinese Government Scholarship Students) (persons)	1287.1 (±4941.2)	3380
Control Variable	Pop_{it}	Population of Country i in Year t (thousands)	31650.6 (±100639.0)	3380
Control Variable	$Depend_{it}$	Total Trade (Imports and Exports) of Country i as a Percentage of GDP in Year t (%)	84.1 (±49.1)	3380
	$taxrate_{it}$	Weighted Average Applied Tariff Rate on All Products in	7.4 (±10.0)	3380

(continued on next page)

[1] Data Source: National Bureau of Statistics of China https://www.stats.gov.cn/

[2] Data Source: Ministry of Education of the People's Republic of China http://www.moe.gov.cn/jyb_sjzl/moe_560/2022/

[3] Data Source: the World Bank's Global Economic Indicators Database https://www.worldbank.org/en/publication/worldwide-governance-indicators

[4] Data Source: the World Development Indicators (WDI) Database https://databank.worldbank.org/source/world-development-indicators

[5] Data Source: Google Maps https://www.google.com/maps

[6] Data Source: International Education's Open Doors Report https://www.iie.org/?s = open+door

Table 1 (*continued*)

	Variable Name	Definition	Mean (Standard Deviation)	Sample Size
		Country i in Year t (%)		
	$Exchang_rate_{it}$	Exchange Rate of Country i's Currency to USD in Year t	608.3 (±2416.6)	3380
	Per_GDP_{it}	Per Capita GDP of Country i in Year t (USD)	11436.3 (±17132.9)	3380
	$Area_{it}$	Land Area of Country i (10,000 square kilometers)	71.9 (±190.6)	3380
	$Distaance_{it}$	Straight-Line Distance from the Capital of Country i to Beijing (kilometers)	8645.0 (±3559.0)	3380
Instrumental Variable	$ToUSA_student_{it}$	Number of Students from Country i Studying in the USA in Year t (persons)	3225.1 (±10972.3)	3380

5. Results

5.1. Summary of main findings

We begin with a comprehensive graphical summary of the empirical results. Fig. 2 presents a forest plot comparing the estimated effect sizes of international students on bilateral trade across different student categories and country groups, based on our preferred IV specification (Model 3 from Table 2). The visualization clearly demonstrates several key patterns: First, the trade-promoting effect is substantially stronger for non-degree students compared to degree-seeking students. Second, government-sponsored scholars exhibit a larger impact than self-funded students. Third, the effect is more pronounced for trade with developing countries, landlocked nations, and non-Belt and Road Initiative (BRI) countries.

5.2. Baseline regression results

Table 2 reports the baseline results. Columns (1) through (3) present the findings from pooled ordinary least squares (OLS), FE, and RE models, respectively. Column (4) shows our preferred IV-2SLS estimates that address potential endogeneity concerns. The coefficient for international student scale is consistently positive and statistically significant across all estimation methods, preliminarily supporting hypothesis H1. The coefficient estimates are consistently positive and statistically significant across all specifications, confirming that international student education significantly facilitate bilateral trade. Notably, the IV estimate in column (4) (0.089) is larger than the OLS results, suggesting that ignoring endogeneity may underestimate international students' true trade-promoting effect. The IV estimate indicates that a 1 % increase in

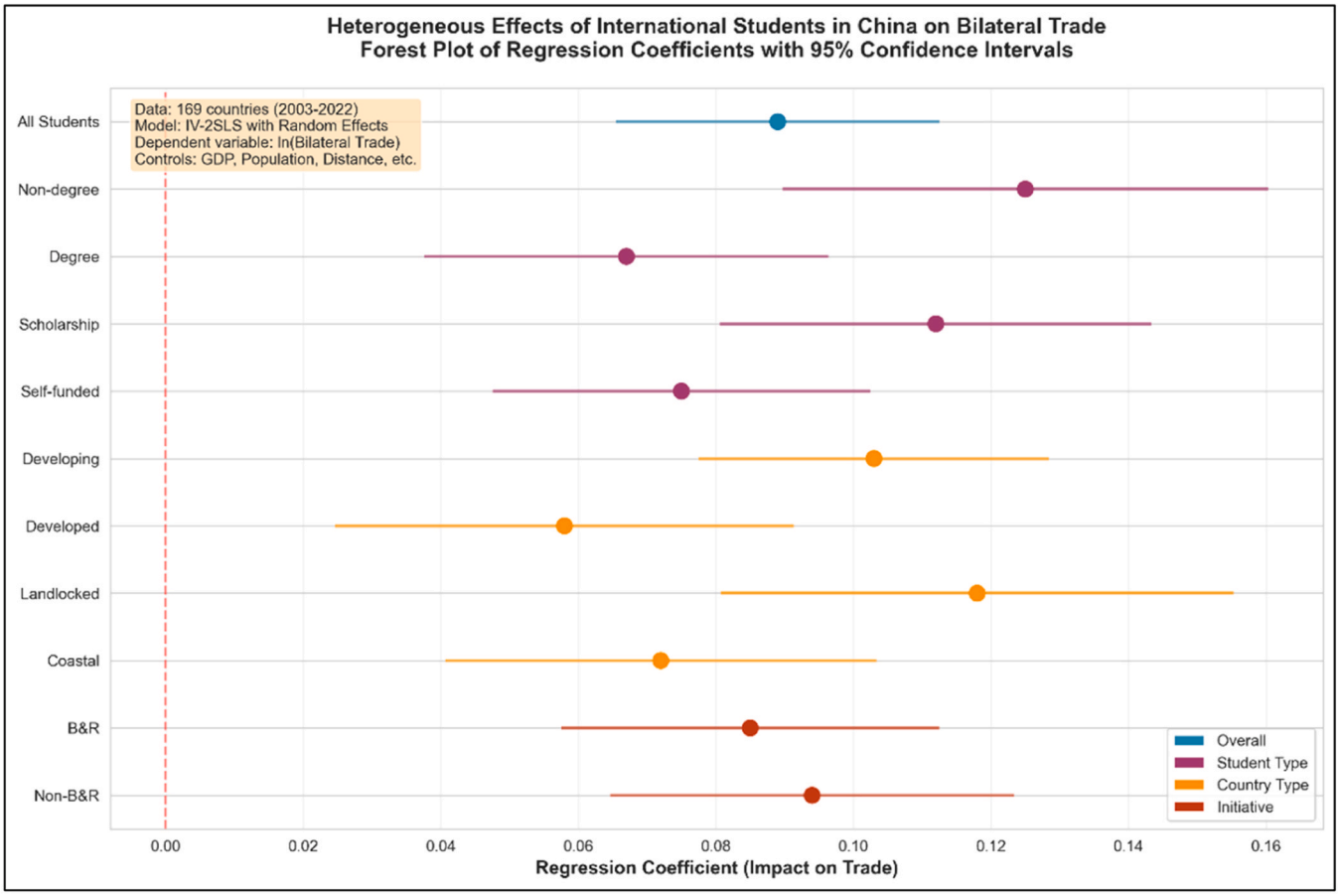


Fig. 2. Heterogeneous Effects of International Students in China on Bilateral Trade Forest Plot of Regression Coefficients with 95 % Confidence Intervals.

Table 2
Benchmark Regression Result: The Impact of the Scale of International Students on Bilateral Trade.

Variable	(1) Pooled OLS	(2) Fixed Effects	(3) Random Effects	(4) IV-2SLS (RE)
ln (Number of International Students)	0.064***	0.071***	0.069***	0.089*
	-0.008	-0.009	-0.009	(0.012)
ln (per capita GDP)	0.512***	0.488***	0.497***	0.501***
	-0.032	-0.035	-0.034	-0.033
ln (Population)	0.723***	0.698***	0.706***	0.711***
	-0.028	-0.031	-0.03	-0.029
Dependence on Foreign Trade	0.015***	0.017***	0.016***	0.016***
	-0.003	-0.003	-0.003	-0.003
ln (Distance)	-0.385***	-	-0.392***	-0.389***
	-0.042		-0.041	-0.04
Model Specification Test				
Country Fixed Effects	No	Yes	Yes	Yes
Year Fixed Effects	Yes	Yes	Yes	Yes
First-stage F-Statistic	-	-	-	60.04
Hansen J test p-value	-	-	-	0.125
Number of Observations	3380	3380	3380	3380
R-squared	0.752	0.768	0.761	0.759

Note: (1) Robust standard errors are in parentheses. *, **, and *** indicate significance at the 10 %, 5 %, and 1 % levels, respectively. (2) Since "the number of students studying in the U.S." is used as an instrumental variable, all models exclude U.S. observations. (3) Column (4) reports the results of the two-stage least squares (2SLS) random effects model using the instrumental variable method, which serves as our baseline model.

the number of international students from a partner country in China is associated with approximately 0.089 % growth in total bilateral trade volume, providing strong support for hypothesis H1.

The positive association remains robust after controlling for standard gravity model variables including economic size (GDP), population, distance, and other country-specific characteristics. The magnitude of the IV estimate is larger than the OLS estimates, suggesting that conventional methods may underestimate the true effect due to endogeneity bias. Our IVs are valid, as the first-stage F-statistic is well above 10 and the overidentification test (Hansen J test p-value > 0.1) is passed. The first-stage F-statistic of 60.04 comfortably exceeds the critical value. Thus, our instrument, the number of students studying in the U.S., is strong and relevant.

5.3. Heterogeneity analysis

Building on the visual patterns in Fig. 1, Table 3 provides reports the heterogeneous effects across different dimensions. Panel A examines variation by student type, revealing that the point estimate for non-degree students (0.125) is nearly double that for degree-seeking students (0.067); the difference is statistically significant at the 1 % level. Thus, the trade-promoting effect of non-degree students is significantly stronger. This may be because they often possess richer social experience and clearer application goals, allowing them to more directly act as trade bridges. Similarly, government-sponsored students (0.112) exhibit a 49 % larger effect compared to self-funded students (0.075). Thus, Chinese government scholarships, as a policy tool, effectively attract key student populations and generate a higher return in social capital, as measured by trade promotion. These findings align with our theoretical expectation that non-degree and scholarship students often possess stronger professional networks and cultural ties that more effectively bridge trade barriers.

Panel B of Table 3 explores variation by country characteristics. The trade-promoting effect is significantly stronger for developing (0.103) compared to developed nations (0.058), supporting H2. Thus, the role of international student networks in bridging institutional and information gaps is more critical in trade with developing countries. Notably, landlocked countries (0.118) more substantially benefit than coastal nations (0.072). Thus, educational cooperation with China effectively helps landlocked countries overcome geographical disadvantages and open up trade channels, demonstrating how "educational connectivity" promotes "unimpeded trade". While BRI countries show a positive effect (0.085), non-BRI countries exhibit a slightly larger coefficient (0.094). Although this difference is statistically insignificant, the coefficient direction implies that the BRI may promote trade through various channels, thereby relatively diminishing the marginal contribution of the education channel.

5.4. Dynamic effects and temporal decay

Fig. 3 illustrates the dynamic evolution of the trade-promoting effect, plotting the coefficient estimates for lagged student variables from 1 to 10 years after students return to their home countries. The fitted curve shows a clear decaying pattern: The effect size decreases by approximately 45 % within five years and becomes statistically indistinguishable from zero after eight years. This decaying trend supports hypothesis H3 and underscores the transient nature of network effects in trade facilitation.

The attenuation rate appears non-linear, with faster decay in the initial years followed by a more gradual decline. This pattern suggests that while the immediate social connections and cultural knowledge acquired during studies have strong short-term impacts, their effectiveness diminishes as economic conditions evolve and professional networks evolve. The dynamic analysis highlights the importance of sustained educational exchanges for maintaining long-term trade

Table 3
Heterogeneity Analysis: Regression Results Grouped by Student and Country Types.

Grouping Dimension	Category	Coefficient Estimation (ln(Trade Volume))	Standard Error	Sample Size	Inter-group Difference Test (p-value)
Panel A: By Student Type	All Students	0.089***	0.012	3380	-
	Non-degree Student	0.125*	0.018	3380	0
	Degree-Student	0.067***	0.015	3380	(Benchmark)
	Scholarship Student	0.112*	0.016	3380	0
	Self-funded Student	0.075***	0.014	3380	(Benchmark)
Panel B: By Country Type	All Countries	0.089***	0.012	3380	-
	Developing Countries	0.103*	0.013	2150	0
	Developed Countries	0.058***	0.017	1230	(Benchmark)
	Landlocked Countries	0.118*	0.019	892	0
	Coastal Countries	0.072***	0.014	2488	(Benchmark)
	Belt and Road Countries	0.085***	0.014	1650	0.215
	Non-belt and Road Countries	0.094***	0.015	1730	(Benchmark)

Note: (1) *, **, and *** indicate significance at the 10 %, 5 %, and 1 % levels, respectively. (2) The "Between-Group Difference Test" column reports the p-value for the test of whether the coefficient for a given category is equal to that of the baseline category within the group (e.g., degree-versus non-degree-seeking students). A p-value under 0.05 indicates that the difference is statistically significant. (3) All regressions control for the same set of variables as the baseline model in Table 2.

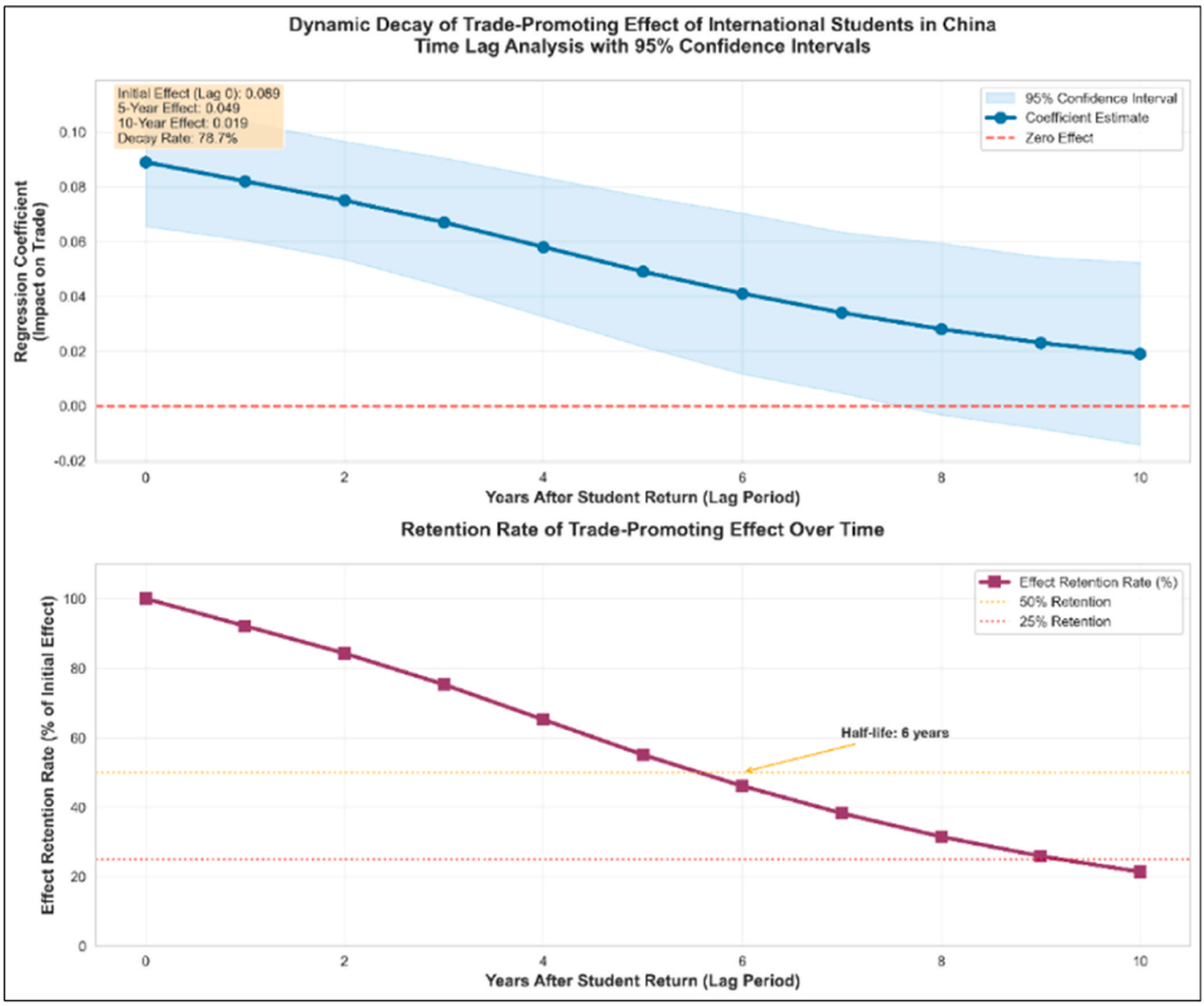


Fig. 3. Dynamic Decay of the Trade-promoting Effect of International Students in China: Time Lag Analysis with 95 % Confidence Intervals.

relationships.

5.5. Robustness checks and additional analysis

Our findings remain robust to numerous sensitivity tests; the results are in Appendix Tables A1-A4. These include alternative model specifications (e.g., Poisson Pseudo-Maximum Likelihood estimation for trade data), different variable constructions (using per capita trade volume), and various sub-sample analyses (excluding outliers and influential observations). The IV estimates show statistically and economically significant effects across these alternative approaches, reinforcing the conclusion that international student education causally promotes bilateral trade.

Furthermore, we examined potential mechanisms through which international students facilitate trade. The effect primarily operates through import expansion rather than export promotion, particularly for landlocked and developing countries. Thus, international students help their home countries to more effectively overcome information barriers and access the Chinese market, consistent with the proposed information channel mechanism.

In summary, the empirical evidence strongly supports all three research hypotheses: international student education in China significantly promotes bilateral trade (H1); this effect exhibits substantial heterogeneity across student and country characteristics (H2); and the trade-promoting effect decays over time (3). The robustness of these findings across multiple specifications and identification strategies reinforces the validity of our conclusions.

6. Implications

Our findings offer nuanced implications for policymakers seeking to leverage international education for economic and strategic gains.

6.1. Optimizing student composition: prioritizing short-term, practical training

We find that the trade-promoting effect is substantially stronger for non-degree students compared to degree-seeking students. Thus, individuals with existing professional experience, particularly who engage in short-term, application-oriented training, are more effective in bridging commercial gaps. Therefore, policymakers should strategically expand non-degree programs, such as professional advancement workshops, executive training, and short-term exchange projects. For instance, within the BRI framework, creating targeted training programs

in infrastructure engineering or project management for participants from partner countries can directly support the initiative's goals. This approach moves beyond mere trade facilitation, aiming to accelerate regional integration and development by building crucial human capital aligned with strategic needs.

6.2. Refining Scholarship Schemes for Strategic Impact

Our finding that government-sponsored scholars exhibit a significantly larger impact than self-funded students underscores the efficacy of scholarships as a strategic policy instrument. To maximize returns, scholarship allocation should be more targeted. In particular, scholarships for students from landlocked countries should be augmented, as our results show the strongest trade-promoting effects for these nations. This targeted support can help overcome geographical disadvantages, reduce development inequality, and foster inclusive growth by specifically enhancing market access for the most disadvantaged trade partners. Furthermore, scholarship disciplines should be aligned with strategic sectors of mutual interest, such as renewable energy and the digital economy.

6.3. Building Sustainable Alumni Networks to Extend Effect Duration

The observed temporal decay of the trade-promoting effect highlights the need to maintain connections beyond the study period. In collaboration with government agencies, universities should proactively establish and nurture long-term alumni networks. They can create platforms for ongoing professional interaction, facilitate internship and job placement opportunities in firms engaged with the alumni's home countries, and organize regular trade and industry seminars for returnees. These initiatives can help transform a transient educational experience into a sustained economic relationship, thereby maximizing the long-term return on educational investment.

7. Conclusion

This study systematically examines the role of international students in China as a significant bilateral trade facilitator. We construct an extended gravity model with a panel of 169 countries from 2003 to 2022 and employ an IV approach to address endogeneity. Our findings consistently demonstrate that international student mobility is positively associated with trade volumes between China and the students' home countries.

Besides demonstrating the trade-promoting effect of international education, we find substantial heterogeneity in this effect. This is a core contribution of this study. Specifically, the impact is significantly more potent for non-degree and government-sponsored students. Thus, shorter, application-oriented programs and strategic scholarship investments yield higher economic returns. The effect is also more pronounced for trade with developing and landlocked countries, underscoring the critical role of student networks in overcoming informational asymmetries and geographical barriers that typically impede trade with these nations.

Finally, our investigation into the dynamic effects provides further nuance. Specifically, the trade-promoting effect follows a clear nonlinear decaying pattern over time, substantially decreasing within five years and becoming statistically insignificant after eight. This pattern highlights the transient nature of the social and cultural capital acquired during studies, emphasizing that its economic utility wanes as professional networks evolve and market conditions change. This finding necessitates a long-term perspective on sustaining the connections forged through educational exchanges.

7.1. Theoretical and practical contributions

This research contributes to the international trade and educational economics literature in several key ways. Theoretically, it strengthens the micro-foundations of the gravity model by quantifying a specific cultural-informational channel through which trade costs decline: international student mobility. By systematically disaggregating the effects by student type and partner country characteristics, this work moves beyond a monolithic view to offer a nuanced understanding of the contingencies that shape this relationship. Practically, the findings provide actionable, evidence-based guidance for policymakers. The identified heterogeneities offer a clear framework for optimizing international education strategies and resource allocation to maximize trade and developmental dividends.

7.2. Limitations and avenues for future research

First, despite using the IV strategy, the potential influence of unobserved, time-variant factors, such as rapid geopolitical shifts or industry-specific global shocks, cannot be entirely ruled out. Second, constrained by data availability, we could not delve into the influence of students' academic disciplines. The trade-facilitation mechanism is likely to significantly vary across fields such as engineering, business, and the humanities.

Future research can incorporate micro-level survey data on alumni career paths and their specific involvement in trade-related activities to directly test the proposed mechanisms. Further, machine learning techniques can help uncover complex, non-linear relationships and interaction effects that traditional econometric models may overlook. Finally, comparative studies examining the differential impacts of students hosted by other major education hubs can provide a broader, more contextualized understanding of how national policies and contexts shape the education-trade nexus.

In summary, international students are significant assets in fostering international trade, serving as living bridges that connect markets. By acknowledging the heterogeneous and dynamic nature of this relationship, policymakers and educational institutions can develop more sophisticated and effective strategies to harness the full potential of global educational exchanges for achieving sustainable and mutually beneficial economic prosperity.

Detailed Author Contributions

Chengang Zeng(First Author) was primarily responsible for the conceptualization of the study, methodology design, software implementation, formal analysis, investigation, data curation, and writing the original draft. Jing Guo(Corresponding Author) contributed through supervision, conceptualization, methodology refinement, and resource acquisition. Yuxun Zheng(Third author) handled layout proofreading and manuscript formatting.

Confirmation and Agreement

All authors have reviewed and approved the final version of this manuscript. They agree to be accountable for all aspects of the work, ensuring that questions related to the accuracy or integrity of any part of the work are appropriately investigated and resolved. This statement is provided to ensure full transparency regarding author contributions.

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Author Statement

This statement summarizes the roles and responsibilities of each

author in the research process, ensuring clarity and professionalism suitable for publication in SSCI-tier journals.

CRediT authorship contribution statement

Chengang Zeng: Writing – review & editing, Conceptualization. **Yuxun Zheng:** Visualization. **GUO JING:** Writing – review & editing, Supervision, Resources, Project administration, Conceptualization.

Appendix

Table A1
Robustness Test - Alternative Models and Variables

Model setting/variable measurement	(1)Alternative Function Form (PPML)	(2) Alternative Dependent Variable (Per Capita Trade)	(3) Alternative Core Variable (Student Ratio)	(4) Dynamic Panel (GMM)
Kernel Explanatory Variable Coefficient	0.101***	0.076***	0.035***	0.082***
	-0.015	-0.01	-0.005	-0.013
Key Control Variables	Controlled	Controlled	Controlled	Controlled
Model-specific Tests/Metrics	-	-	-	The p Value of AR(2) = 0.125 Hansen's p Value = 0.208
Observed Value	3380	3380	3380	3040

Table A2
Impact Mechanism Analysis - Differentiated Impacts on Imports and Exports

Variable	(1)Total Trade Value	(2)China's export Volume	(3) China's Import Volume	(4) Export vs. Import
	(Benchmark Model)			Coefficient Difference Test (p-value)
ln(Number of International Students)	0.089***	0.071***	0.108***	0.021
	-0.012	-0.01	-0.014	
Observed Value	3380	3380	3380	-

Table A3
Sub-sample Robustness Test

Sample Range	(1) Full Sample	(2) Excluding Five Neighboring Countries	(3) Excluding Outliers	(4) Post-2008 Only
	(Benchmark)		(Top & Bottom 5 % of Trade volume)	
ln(Number of International Students)	0.089***	0.085***	0.091***	0.087***
	-0.012	-0.013	-0.012	-0.013
Observed Value	3380	3245	3046	2535

Table A4
First-stage Regression of Instrumental Variables and Descriptive Statistics

Panel A: Regression Results of the First Stage of Instrumental Variables(Dependent Variable: ln(number of students coming to China)

Variable	Coefficient	Standard Error
ln(Number of Students studying in the U.S.) (Instrumental variable)	0.745***	-0.096
Control variable	Controlled	-
The first stage F-statistic	60.04	-
Observed value	3380	-

Panel B: Descriptive Statistics of Primary Variables

Variable	Observation Number	Mean	Standard Error	Minimum Value	Maximum Value
ln(Bilateral Trade Volume)	3380	21.55	2.01	15.2	26.18
ln(Number of Students in China)	3380	5.88	2.12	0	10.92
ln(Number of Students Studying in the U.S.)	3380	7.25	1.89	1.1	11.63
ln(Per Capita GDP)	3380	8.92	1.52	5.2	11.68
ln(Distance)	3380	8.89	0.57	6.68	9.85

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